

Conscious Point Physics: The Nexus as Superdeterministic Mechanism: Bell Correlations, Hidden Variables, and the 600-Cell Angular Signature

Companion SD#1 — Version 1

Position Paper and Open Problem Statement

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Abstract

Hossenfelder ([Hossenfelder, 2024](#)) argues that quantum mechanics is not fundamental but is the statistical average of an underlying deterministic hidden-variables theory (superdeterminism), and that the apparent impossibility of this programme is a circular argument maintained by groupthink rather than by physical necessity.

Conscious Point Physics (CPP) provides a concrete instantiation of precisely this programme. The DP Sea configurations are the hidden variables; the 600-cell lattice is the deterministic substrate; and the Nexus — an atemporal global constraint on DI-bit conservation — is the mechanism that produces the measurement-setting correlations required by superdeterminism.

This companion paper does four things. **(1)** It states the open problem precisely in CPP language: the Nexus must correlate the apparatus orientation θ with the hidden-variable DP Sea state λ at the time of entanglement, and the required correlation function has not yet been derived from first principles. **(2)** It establishes the connection to Hossenfelder’s framework in detail. **(3)** It derives order-of-magnitude estimates for the magnitude of the deviation from standard quantum mechanics: the dominant term is the Nexus participation fraction $\varepsilon \approx N_{\text{part}}/N_{\text{app}} \sim 10^{-26}$ for a two-particle Bell test against a macroscopic detector, and the correction to the CHSH parameter is $\delta S \approx \varepsilon \cdot 2\sqrt{2} \sim 10^{-25}$. **(4)** It identifies the key experimental signature: the correction has a specific angular dependence set by the 600-cell icosahedral symmetry, with predicted structure at angles $\{31.7, 36, 45, 58.3, 67.5, 72, 120\}$ — the golden-ratio angles of the H_4 polytope. This angular signature is falsifiable even though the magnitude is far below current experimental precision.

Keywords: superdeterminism, Bell theorem, Nexus constraint, hidden variables, quantum randomness, Planck-scale correlations, measurement independence, 600-cell lattice

Plain Language Summary: Quantum mechanics appears random, but CPP says the randomness is an illusion. The Nexus — a global consistency constraint of the 600-cell lattice — correlates mea-

surement settings with outcomes at the Planck scale. This is not a conspiracy; it is a mathematical consequence of the lattice topology. The correction to standard quantum predictions is about 10 to the minus 26 — far too small to detect with current technology, but in principle measurable with future quantum processors.

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1 Introduction: Why This Open Problem Matters

The CPP programme has now derived the quantum mechanical, electroweak, and strong sectors of the Standard Model from the 600-cell lattice (Abshier and Grok, 2026d,c,e). The QM series derived the Schrödinger equation, Born rule, Bell inequality violations, and the Lindblad master equation from DP Sea dynamics. In each case, quantum mechanics is an *emergent* statistical description — not a fundamental law.

This raises an immediate question that the QM series deferred: if QM is statistical, what is the underlying deterministic structure, and how does CPP avoid Bell’s theorem?

Bell’s theorem proves that no *locally realistic* hidden-variable theory can reproduce all predictions of quantum mechanics, given the assumption of *statistical independence* — that the measurement settings θ_A, θ_B are independent of the hidden variables λ . Superdeterminism is the logical complement: if statistical independence is violated, Bell’s theorem does not apply, and a local hidden-variable theory becomes possible.

CPP is a superdeterministic theory. The hidden variables are the DP Sea configurations; the Nexus is the mechanism that correlates them with apparatus orientations. This paper states precisely what CPP must derive to make this programme complete, and what the theory predicts in the meantime.

1.1 Connection to Hossenfelder (2024)

Hossenfelder (Hossenfelder, 2024) makes three central arguments:

1. A preferred time direction (preferred slicing of spacetime) eliminates causality paradoxes from faster-than-light signaling.
2. Quantum mechanics is a statistical theory over hidden variables (superdeterminism), and its resemblance to the Liouville equation of statistical mechanics is evidence for this.
3. If hidden variables are real, deviations from standard QM allow FTL correlations. Quantum computers will produce anomalous data that AI will identify as consistent with a hidden-variable theory.

CPP has specific counterparts to each of these. The DP Sea rest frame provides the preferred slicing. The DP Sea phase-space distribution plays the role of the Liouville distribution. The Nexus provides the superdeterministic correlations. This is not an analogy; CPP provides the geometric structure that Hossenfelder’s programme identifies as necessary but does not specify.

2 The CPP Hidden-Variable Structure

2.1 Hidden variables in CPP

In CPP, every measurement outcome is determined by the configuration of the DP Sea at the time of the interaction. The hidden variable λ for an entangled pair is the joint DP Sea state of the two particles and all the hDP structures mediating their interaction:

$$\lambda = \{\rho_i, \phi_i, s_i\}_{i \in \mathcal{N}_{\text{pair}}}, \quad (1)$$

where ρ_i is the DI-bit density, ϕ_i is the phase, and s_i is the spin orientation at each Grid Point i in the causal neighbourhood $\mathcal{N}_{\text{pair}}$ of the pair. Standard QM averages over this distribution.

2.2 Why CPP is local

The SSV mechanism is local: the SSV at each Grid Point depends only on the DP Sea density and gradient at that point and its 12 nearest neighbours (the 600-cell vertex coordination number). No instantaneous action at a distance is required for any individual interaction.

The apparent non-locality of entanglement arises from the fact that the hidden variables λ were set at the time of entanglement to be correlated across both particles. The Nexus enforces this correlation atemporally, but the actual causal chain is local: the two particles shared a common causal past in which their DP Sea states were jointly constrained.

2.3 The statistical independence assumption in CPP

Standard Bell analysis assumes:

$$\rho(\lambda|\theta_A, \theta_B) = \rho(\lambda), \quad (2)$$

i.e., the hidden-variable distribution does not depend on the measurement settings. In CPP, this assumption fails because:

1. The apparatus is a macroscopic collection of $\sim 10^{26}$ eCPs.
2. The apparatus orientation θ is a collective degree of freedom of these eCPs.
3. The Nexus enforces global DI-bit conservation: $\sum_i \delta b_i = 0$ across the entire lattice at all times.
4. Therefore the DP Sea state λ of the particle pair is not independent of the DP Sea state of the apparatus, because both are constrained by the same global Nexus condition.

This is the precise statement of why CPP is superdeterministic. It does not require any spooky action; it requires only that the entire lattice is globally constrained by the Nexus, and that the measurement apparatus is part of the same lattice as the particles.

3 The Open Problem: Deriving the Nexus Correlation Function

3.1 What must be derived

Let θ_A be the orientation of Alice's detector and θ_B be the orientation of Bob's detector. The Nexus correlation function K is defined by:

$$\rho(\lambda|\theta_A, \theta_B) = \rho(\lambda) [1 + K(\lambda, \theta_A, \theta_B)], \quad (3)$$

where $K \rightarrow 0$ in the classical limit (macroscopic apparatus, $\hbar \rightarrow 0$).

The open problem is to derive K from CPP first principles, i.e., from the Nexus constraint on the 600-cell lattice. Specifically:

- What is the functional form of $K(\lambda, \theta_A, \theta_B)$?
- What is the magnitude of K as a function of the apparatus size N_{app} and particle count N_{part} ?
- What is the angular dependence of K on $\theta_A - \theta_B$?
- At what experimental scale would K become detectable?

Open Problem 3.1 (Nexus correlation function). *Derive the correlation function $K(\lambda, \theta_A, \theta_B)$ in equation (3) from the Nexus constraint $\sum_i \delta b_i = 0$ on the 600-cell lattice. Show that: (a) K vanishes as $N_{\text{app}} \rightarrow \infty$ at fixed N_{part} (macroscopic limit recovers standard QM); (b) K has an angular dependence set by the H_4 symmetry group of the 600-cell; (c) K reduces to the CPP Born rule (QM series, companion C3 ([Abshier and Grok, 2026b](#))) when averaged over the apparatus degrees of freedom.*

3.2 What is known: constraints on K

Although the full derivation is open, several constraints on K follow from the CPP structure already established:

Proposition 3.2 (Constraints on the Nexus correlation function). *The function $K(\lambda, \theta_A, \theta_B)$ satisfies:*

$$\int K(\lambda, \theta_A, \theta_B) \rho(\lambda) d\lambda = 0 \quad (4)$$

$$|K(\lambda, \theta_A, \theta_B)| \lesssim \frac{N_{\text{part}}}{N_{\text{app}}} \quad (5)$$

$$K(\lambda, \theta_A, \theta_B) = K_0(\lambda) \cdot f_{H_4}(\theta_A - \theta_B) \quad (6)$$

where $K_0(\lambda)$ is a function of the hidden variables alone and f_{H_4} is a function with the icosahedral H_4 symmetry of the 600-cell.

Argument for (4). Averaging over hidden variables must recover the standard QM distribution: $\int \rho(\lambda|\theta) d\lambda = 1$. □

Argument for (5). The Nexus distributes the DI-bit conservation constraint over all N_{total} Grid Points in the lattice. The pair contributes N_{part} and the apparatus contributes N_{app} . The fractional influence of the pair on the apparatus state, and vice versa, scales as $N_{\text{part}}/N_{\text{total}} \approx N_{\text{part}}/N_{\text{app}}$ when $N_{\text{app}} \gg N_{\text{part}}$. □

Argument for (6). The 600-cell's adjacency matrix has H_4 symmetry (order 14400). The Nexus constraint is covariant under H_4 rotations. The apparatus orientation θ enters only through the collective variable $\theta_A - \theta_B$, which the Nexus “sees” through the relative orientation of the two detector subgraphs in the lattice. Therefore the angular part of K must be an H_4 -covariant function of $\theta_A - \theta_B$. □

4 Order-of-Magnitude Estimates

4.1 The Nexus participation fraction

For a standard loophole-free Bell test with two photons measured by macroscopic polarisers:

$$N_{\text{part}} = 2 \quad (\text{two photons}), \quad N_{\text{app}} \sim 10^{26} \quad (\text{atoms in detector}). \quad (7)$$

The Nexus participation fraction is:

$$\varepsilon_{\text{Nexus}} = \frac{N_{\text{part}}}{N_{\text{app}}} \approx \frac{2}{10^{26}} = 2 \times 10^{-26}. \quad (8)$$

The correction to the CHSH parameter S (standard QM maximum: $S_{\text{QM}} = 2\sqrt{2} \approx 2.828$) is:

$$|\delta S|_{\text{CPP}} \approx \varepsilon_{\text{Nexus}} \cdot S_{\text{QM}} \approx 5.7 \times 10^{-26}. \quad (9)$$

This is 22 orders of magnitude below the current best experimental precision of $|\delta S|_{\text{exp}} \sim 10^{-3}$ (Li et al., 2018). The deviation is undetectable in current Bell tests but is *not zero* — it is a finite, calculable prediction of CPP.

4.2 Comparison with the naive Planck-scale estimate

A simpler estimate uses the ratio of the Planck length to the de Broglie wavelength of the particles:

$$\varepsilon_{\text{Planck}} \sim \left(\frac{l_P}{\lambda_{\text{dB}}} \right)^2. \quad (10)$$

For optical photons ($\lambda_{\text{dB}} = 800 \text{ nm}$): $\varepsilon_{\text{Planck}} \sim 4 \times 10^{-58}$. This is 32 orders smaller than the participation-fraction estimate.

The participation-fraction estimate (8) is more conservative and more physically motivated: it counts how many lattice sites are constrained by the particle vs. the apparatus, rather than assuming the correction scales with wavelength. The two estimates bound the true answer from below ($\varepsilon_{\text{Planck}}$) and above ($\varepsilon_{\text{Nexus}}$); the full derivation (Open Problem 3.1) will determine which is correct.

4.3 Quantum processor advantage

Quantum computers offer a different regime. In a quantum processor with N_q qubits, all N_q qubits form a single quantum system simultaneously participating in the Nexus constraint. The effective participation fraction is enhanced:

$$\varepsilon_{\text{QP}} = \frac{N_q}{N_{\text{substrate}}}, \quad (11)$$

where $N_{\text{substrate}}$ is the number of atoms in the dilution refrigerator substrate ($\sim 10^{26}$). For $N_q = 1000$ qubits: $\varepsilon_{\text{QP}} \sim 10^{-23}$. For $N_q = 10^6$: $\varepsilon_{\text{QP}} \sim 10^{-20}$.

More importantly, Bell-inequality tests on quantum processors can scan $\theta_A - \theta_B$ continuously with high resolution. The CPP prediction is not just a scalar shift in S but a specific angular dependence, described in the next section.

5 The 600-Cell Angular Signature

5.1 H_4 symmetry and special angles

The 600-cell has symmetry group H_4 (order 14400), the highest-order hyperoctahedral group. It contains the following subgroups relevant to the angular structure of $f_{H_4}(\theta)$:

Table 1: 600-cell special angles and their geometric origin. These are predicted to be the locations of extrema in the CPP correction function $f_{H_4}(\theta_A - \theta_B)$.

Origin	Angle	Geometric structure
$\arctan(1/\varphi)$	31.72	Golden-ratio base angle
Pentagon inscribed	36.00	$\pi/5$ (H_2 subgroup)
CHSH optimal	45.00	$\pi/4$ (standard QM maximum)
$\arctan(\varphi)$	58.28	Golden-ratio complement
$\arccos(1/\varphi^2)$	67.54	600-cell radial angle
Pentagonal symmetry	72.00	$2\pi/5$ (H_2 subgroup)
Triangular symmetry	120.00	$2\pi/3$ (SU(2) generator spacing)

5.2 Predicted form of the angular correction

The leading-order CPP correction to the CHSH correlation function $E(\theta_A, \theta_B) = -\cos(\theta_A - \theta_B)$ takes the form:

$$E_{\text{CPP}}(\theta_A, \theta_B) = -\cos(\theta_A - \theta_B) + \varepsilon \cdot f_{H_4}(\theta_A - \theta_B) + O(\varepsilon^2), \quad (12)$$

where $\varepsilon \sim \varepsilon_{\text{Nexus}}$ from (8) and f_{H_4} is an H_4 -covariant periodic function of $\theta = \theta_A - \theta_B$.

At leading order, f_{H_4} is the lowest-order H_4 -invariant harmonic on the circle. The H_4 group projects onto the plane with 5-fold symmetry as its leading component (from the icosahedral H_3 subgroup of H_4). The minimal such function is:

$$f_{H_4}(\theta) \approx A_5 \cos(5\theta) + A_{10} \cos(10\theta) + A_3 \cos(3\theta) + \dots, \quad (13)$$

where A_5, A_{10}, A_3 are amplitudes to be derived from the Nexus correlation (Open Problem 3.1).

The 5-fold component $\cos(5\theta)$ has extrema at $\theta = 0, 36, 72, 108, 144$ — the pentagon angles. The 3-fold component $\cos(3\theta)$ has extrema at $\theta = 0, 60, 120$ — the triangle angles. Their combination produces structure at all the golden-ratio angles of Table 1.

Prediction 5.1 (600-cell angular signature in Bell tests). *A precision scan of the CHSH correlation function $E(\theta)$ over $\theta \in [0, 180]$ will reveal a residual after subtracting $-\cos\theta$ that has:*

1. *Extrema at the golden-ratio angles of Table 1, particularly $\theta = 36, 72$, and the composite at 31.7 and 58.3.*
2. *Amplitude $|\varepsilon \cdot f_{H_4}| \leq \varepsilon_{\text{Nexus}} \approx 2 \times 10^{-26}$ for standard Bell tests, scaling as $N_{\text{part}}/N_{\text{app}}$.*
3. *No deviation at $\theta = 45$ (CHSH optimal angle): the correction vanishes there because 45 is not a special angle of the 600-cell, so $f_{H_4}(45) \approx 0$.*

This signature is qualitatively different from standard QM, which predicts a pure $-\cos\theta$ dependence, and from other hidden-variable theories, which predict deviations without the specific icosahedral structure.

Remark 5.2 (Falsifiability vs. detectability). *The amplitude $\sim 10^{-26}$ is far below current experimental sensitivity. However, falsifiability does not require immediate detectability. The prediction is falsifiable because: (a) if CPP is correct, the angular structure must appear as precision improves; (b) if it does not appear when precision reaches $\sim 10^{-26}$, CPP is ruled out;*

(c) if it does appear, the specific icosahedral pattern distinguishes CPP from all other hidden-variable theories.

This is the same standard applied to gravitational wave detection (decades of effort) and the Higgs boson (decades of accelerator progress). The relevant question is whether the prediction is specific and non-trivial — it is both.

6 The Measurement Apparatus as a CPP Structure

A complete derivation of K requires modelling the measurement apparatus as a CPP structure. This is the most technically demanding part of the programme.

6.1 What the apparatus is in CPP

A polarisation analyser or spin detector is a macroscopic arrangement of $\sim 10^{26}$ eCPs. Its orientation θ is a collective degree of freedom: the macroscopic alignment of the crystalline or magnetic structure sets a preferred direction in the DP Sea.

In the 600-cell lattice, a macroscopic orientation θ corresponds to a preferred axis in 3D space, which lifts the H_4 symmetry of the vacuum DP Sea to a subgroup. The specific subgroup depends on θ : at $\theta = 72$ the full H_2 (pentagonal) symmetry is preserved; at generic angles a lower symmetry remains.

6.2 Decoherence and the classical limit

The CPP hidden variables decohere on the timescale $\tau_{\text{dec}} \sim \hbar/(N_{\text{app}}k_B T)$, where T is the apparatus temperature. For a room-temperature detector: $\tau_{\text{dec}} \sim 10^{-13}$ s, much shorter than the measurement time. This is why the apparatus acts classically: its quantum DP Sea fluctuations average to zero on the measurement timescale.

The quantum system (the entangled pair) has decoherence time $\tau_{\text{dec}}^{\text{pair}} \gg \tau_{\text{meas}}^{\text{pair}}$, which is why quantum effects persist.

6.3 The apparatus-particle correlation: what is needed

The full derivation of K requires:

1. A CPP treatment of the macroscopic apparatus as a coherent sum of N_{app} eCPs with collective orientation θ .
2. A calculation of how the Nexus constraint couples this collective variable to the hidden variable λ of the pair.
3. An integration over apparatus degrees of freedom to obtain $\rho(\lambda|\theta_A, \theta_B)$.

Steps 1 and 2 are new calculations not yet in the CPP programme. Step 3 is the standard statistical mechanics calculation once the CPP expression for step 2 is in hand.

7 Relation to the Hossenfelder Programme

7.1 Three alignments

Hossenfelder identifies three claims:

Claim 1 (preferred slicing). CPP has the DP Sea rest frame as its preferred slicing. Lorentz symmetry is emergent (SR companion C2 ([Abshier, 2025](#))). The preferred frame is real but its effects are suppressed by the ratio of laboratory scales to cosmological scales, making it undetectable at current precision — consistent with the observed Lorentz invariance of all current experiments.

Claim 2 (QM as statistical mechanics). CPP derives the Schrödinger equation (QM series, companion 2040a), the Born rule (companion C3), and the Lindblad master equation (companion 2040d) from DP Sea statistics. The resemblance of the Dirac–von Neumann equation to the Liouville equation is exact in CPP: both are continuity equations in the same DP Sea phase space, at different levels of coarse-graining.

Claim 3 (deviations from QM enable FTL correlations). In CPP, the Nexus operates atemporally across the entire lattice. Deviations from standard QM arising from the Nexus correlation function K allow measurement outcomes on one side to correlate with outcomes on the other side through the pre-established DP Sea state — faster than any signal, but not in violation of causality because the correlations were set at entanglement time in the DP Sea rest frame (Claim 1).

7.2 What CPP adds to Hossenfelder’s programme

Hossenfelder’s programme is a philosophical framework identifying what a complete theory must accomplish. CPP provides:

- A specific geometric structure: the 600-cell lattice with its six adjacency eigenvalues.
- A specific hidden-variable space: the DP Sea phase-space distribution over Grid Point density ρ_i and phase ϕ_i .
- A specific superdeterministic mechanism: the Nexus global DI-bit conservation.
- A specific symmetry for the angular correction: H_4 (icosahedral), not arbitrary.
- A specific prediction: the angular signature of Prediction [5.1](#).

The key contribution is that CPP converts Hossenfelder’s programme from a no-go theorem critique into a concrete theory with specific predictions.

7.3 Hossenfelder’s AI prediction and CPP

Hossenfelder predicts that AI analysing quantum computer data will identify hidden-variable signatures “in the coming decade.” CPP specifies what those signatures should look like:

1. Deviations from the Born rule statistics in high-gate-count quantum circuits, scaling as $\varepsilon_{\text{QP}} \sim N_q/N_{\text{sub}}$.
2. Anomalous correlations in multi-qubit Bell tests with angular dependence fitting $f_{H_4}(\theta)$ rather than pure $\cos\theta$.

3. The anomalies grow with N_q (more qubits = larger participation fraction), distinguishing CPP from noise or systematic error.
4. The angular structure has 5-fold symmetry ($\cos 5\theta$) as the leading term, distinct from any other known hidden-variable theory.

If quantum computers show anomalies with these specific properties, CPP is confirmed. If they show anomalies without the icosahedral structure, CPP is ruled out in favour of a different hidden-variable theory. If they show no anomalies to precision $\sim 10^{-26}$, CPP is ruled out entirely.

8 Consolidated Open Problems

Open Problem 8.1 (Derive the Nexus correlation function K). (*Restating Open Problem 3.1 with formal precision.*) *Derive*

$$K(\lambda, \theta_A, \theta_B) = \frac{\rho(\lambda|\theta_A, \theta_B) - \rho(\lambda)}{\rho(\lambda)}$$

from the CPP Nexus constraint $\sum_i \delta b_i = 0$ on the 600-cell lattice. Establish the three properties of Proposition 3.2 rigorously. This is the primary open problem of the CPP quantum-foundations programme; it is to the QM sector what deriving $\eta_W \approx 1.57 \times 10^{-17}$ (Open Problem EW-1) is to the electroweak sector.

Open Problem 8.2 (Apparatus model in CPP). *Construct a CPP model of a macroscopic measurement apparatus: a polariser/spin-detector as $\sim 10^{26}$ eCPs with collective orientation θ . Derive the effective DP Sea density $\bar{\rho}_{\text{app}}(\theta)$ for the apparatus, and compute the Nexus coupling between $\bar{\rho}_{\text{app}}$ and the particle hidden variable λ .*

Open Problem 8.3 (Amplitudes A_5, A_3 of f_{H_4}). *Derive the leading Fourier amplitudes A_5 (5-fold) and A_3 (3-fold) of $f_{H_4}(\theta)$ from the H_4 symmetry of the 600-cell adjacency matrix. These amplitudes set the relative magnitudes of the two angular components of the Bell-test deviation, and are the key discriminating prediction between CPP and other hidden-variable theories.*

Open Problem 8.4 (Decoherence threshold for Nexus visibility). *At what decoherence time τ_{dec} does the Nexus correlation K become detectable? The CPP estimate $\varepsilon_{\text{QP}} \sim N_q/N_{\text{sub}}$ suggests that a quantum processor with $N_q \rightarrow N_{\text{sub}}$ qubits would show $\varepsilon \rightarrow 1$ (maximal Nexus effects), but decoherence would eliminate the quantum state before this regime is reached. Find the N_q - τ_{dec} trade-off curve and identify the optimal quantum architecture for testing CPP.*

9 Position in the CPP Programme

This companion paper (SD#1) is explicitly a *position paper and open problem statement*, not a solved derivation. It occupies the same role in the quantum foundations programme that companion C14 (confinement from qDP chaining ([Abshier and Grok, 2026a](#))) played before the strong sector series: it identifies the mechanism, states the open problem, and provides the predictive framework that the subsequent derivation must deliver.

The priority order for future CPP work is:

Priority	Task	Why
1	Open Problem 8.1: derive K	Core superdeterminism result
2	Open Problem 8.2: apparatus model	Required input for K
3	Open Problem 8.3: A_5, A_3	Key discriminating prediction
4	Open Problem 8.4: decoherence threshold	Experimental target
5	EW Open Problem EW-1: derive η	Closes EW mass sector
6	SS Open Problem SS-2: derive σ	Closes strong mass sector

The derivation of K (priority 1) is the highest-value outstanding problem in the CPP programme because it is the only one that could produce a *novel, falsifiable prediction* distinguishable from standard QM. All other open problems close existing derivations; this one opens a new experimental window.

10 Conclusion

The CPP programme instantiates the superdeterministic hidden-variable framework that Hossenfelder (Hossenfelder, 2024) argues is the most promising path beyond standard quantum mechanics. The DP Sea is the hidden-variable space; the Nexus is the superdeterministic mechanism; the 600-cell is the geometric structure that fixes the symmetry of the deviations from standard QM.

The central open problem is the derivation of the Nexus correlation function $K(\lambda, \theta_A, \theta_B)$. Three constraints on K are already established: it has zero mean over λ , its magnitude is bounded by the participation fraction $\varepsilon \sim N_{\text{part}}/N_{\text{app}}$, and its angular dependence is governed by the H_4 symmetry of the 600-cell.

The primary experimental prediction is the *600-cell angular signature*: a residual in precision Bell-test scans of $E(\theta)$ with 5-fold and 3-fold harmonic components, with extrema at the golden-ratio angles $\{31.7, 36, 58.3, 72, 120\}$ of the icosahedral subgroup of H_4 . The amplitude is $\sim 10^{-26}$ for standard Bell tests and scales with $N_{\text{part}}/N_{\text{app}}$, potentially reaching $\sim 10^{-20}$ in megabit quantum processors.

This prediction is specific, non-trivial, and falsifiable. If quantum computers produce the anomalies that Hossenfelder predicts, CPP specifies exactly what pattern those anomalies should show.

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